DISTANCE TO NEAREST SHORELINE ----------

Calculate distance of each detection (each row in GISCALC_HUWH_STX2025) to nearest shoreline

Near (Analysis Tools):

- Input Features: GISCALC_HUWH_STX2025
- Near Features: CARIBBEAN_SHORELINES
- Search Radius: Leave blank (and leave default Meters)
- ✅ Location (adds NEAR_X and NEAR_Y cols with coordinates of nearest shoreline)
- Method: Planar
- Field Names: Leave as default
- Distance Unit: Meters

Create and populate 'nearest_shoreline' column in GISCALC_HUWH_STX2025 Attribute Table with full names of nearest shoreline using numeric codes in NEAR_FID column

Attribute Table CARIBBEAN_SHORELINES -> **make note of names** for rows where 'OBJECTID' column matches numbers in GISCALC_HUWH_STX2025 'NEAR_FID' column

Calculate Field (Data Management Tools):

- Input Table: GISCALC_HUWH_STX2025
- Field Name (Existing or New): nearest_shoreline
- nearest_shoreline =

```
get_name(!NEAR_FID!)
```

- Code Block:

```
def get_name(fid):
    if fid == 847:
        return "Saint Croix, U.S. Virgin Islands"
    elif fid == 850:
        return "Buck Island, , U.S. Virgin Islands"
    < etc… >
    else:
        return Non
```

RASTERDEPTH AND RASTERSLOPE ----------

Add new RASTERDEPTH and RASTERSLOPE cols to GISCALC_HUWH_STX2025 Attribute Table with bathymetric depth and slope values in meters/degrees derived from GBC and USC bathy and slope rasters

Extract Multi Values to Points (Spatial Analyst Tools):

- Input Point Features: GISCALC_HUWH_STX2025
- Input Rasters: Output Field Name:
  - GBC15a_bathy_STX2025 RASTERDEPTH_GBC15a

GBC15a_slope_STX2025	RASTERSLOPE_GBC15a
USC10m_bathy_STX2025	RASTERDEPTH_USC10m
USC10m_slope_STX2025	RASTERSLOPE_USC10m

- ✅ Bilinear interpolation of values at point locations

Delete data from 'RASTERDEPTH_GBC15a' and 'RASTERSLOPE_GBC15a' columns in rows where associated USC10m depth/slope column has valid value

Calculate Field (Data Management Tools):

- Input Table: GISCALC_HUWH_STX2025
- Field Name (Existing or New): RASTERDEPTH_GBC15a
- RASTERDEPTH_GBC15a =

```
update(!RASTERDEPTH_USC10m!, !RASTERDEPTH_GBC15a!)
```

- Code Block:

```
def update(usc, gbc):
    if usc is not None:
        return None
    else:
        return gbc
```

Calculate Field (Data Management Tools):

- Input Table: GISCALC_HUWH_STX2025
- Field Name: RASTERSLOPE_GBC15a
- RASTERSLOPE_GBC15a =

```
update(!RASTERSLOPE_USC10m!, !RASTERSLOPE_GBC15a!)
```

- Code Block:

```
def update(usc, gbc):
    if usc is not None:
        return None
    else:
        return gbc
```

Add new clean columns for raster depth, raster slope, and raster resolution. Populate depth and slope columns with USC10m values for rows where data is available and GBC15a values for rows where 10-m res. data is not available. Populate raster resolution column with resolution of raster used to populate depth and slope columns for that row.

Calculate Field (Data Management Tools):

- Input Table: GISCALC_HUWH_STX2025
- Field Name (Existing or New): RASTERDEPTH_m
- RASTERDEPTH_m =

```
get_depth(!RASTERDEPTH_USC10m!, !RASTERDEPTH_GBC15a!)
```

- Code Block:

```
def get_depth(usc, gbc):
    if usc is not None:
        return usc
    elif gbc is not None:
        return gbc
```

```
        else:
            return None
```

Calculate Field (Data Management Tools):

- Input Table: GISCALC_HUWH_STX2025
- Field Name (Existing or New): RASTERSLOPE_deg
- RASTERSLOPE_deg =
  ```
  get_slope(!RASTERSLOPE_USC10m!, !RASTERSLOPE_GBC15a!)
  ```
- Code Block:
  ```
  def get_slope(usc, gbc):
      if usc is not None:
          return usc
      elif gbc is not None:
          return gbc
      else:
          return None
  ```

Calculate Field (Data Management Tools):

- Input Table: GISCALC_HUWH_STX2025
- Field Name (Existing or New): raster_resolution
- raster_resolution =
  ```
  get_res(!RASTERDEPTH_USC10m!, !RASTERDEPTH_GBC15a!)
  ```
- Code Block:
  ```
  def get_res(usc, gbc):
      if usc is not None:
          return "10 meter"
      elif gbc is not None:
          return "15 arc second"
      else:
          return None
  ```

Clean up columns (reorder and rename + delete unnecessary columns)

Attribute Table GISCALC_HUWH_STX2025 -> **Fields Editor:**

- Delete columns: RASTERDEPTH_GBC15a, RASTERSLOPE_GBC15a, RASTERDEPTH_USC10m, RASTERSLOPE_USC10m
- Rename column: NEAR_DIST to NEAR_DIST_m
- Reorder columns: All original cols (ending with oxygen), RASTERDEPTH_m, RASTERSLOPE_deg, raster_resolution, NEAR_DIST_m, nearest_shoreline, NEAR_FID, NEAR_X, NEAR_Y

EXPORT CSV WITH FULL DATA FOR DETECTIONS ----------

Export GISCALC_HUWH_STX2025 as point feature class to mission geodatabase for safekeeping

Right-click layer in Map Contents pane GISCALC_HUWH_STX2025 **-> Data -> Export Features:**

- Input Features: GISCALC_HUWH_STX2025
- Output Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\GISCALC_HUWH_STX2025

Export GISCALC_HUWH_STX2025 data as CSV for main calculations!

Right-click layer in Map Contents pane GISCALC_HUWH_STX2025 **-> Data -> Export Table:**

- Input Table: GISCALC_HUWH_STX2025
- Output Table: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\Output\GISCALC_HUWH_STX2025.csv (*must save OUTSIDE of a gdb to get CSV format)